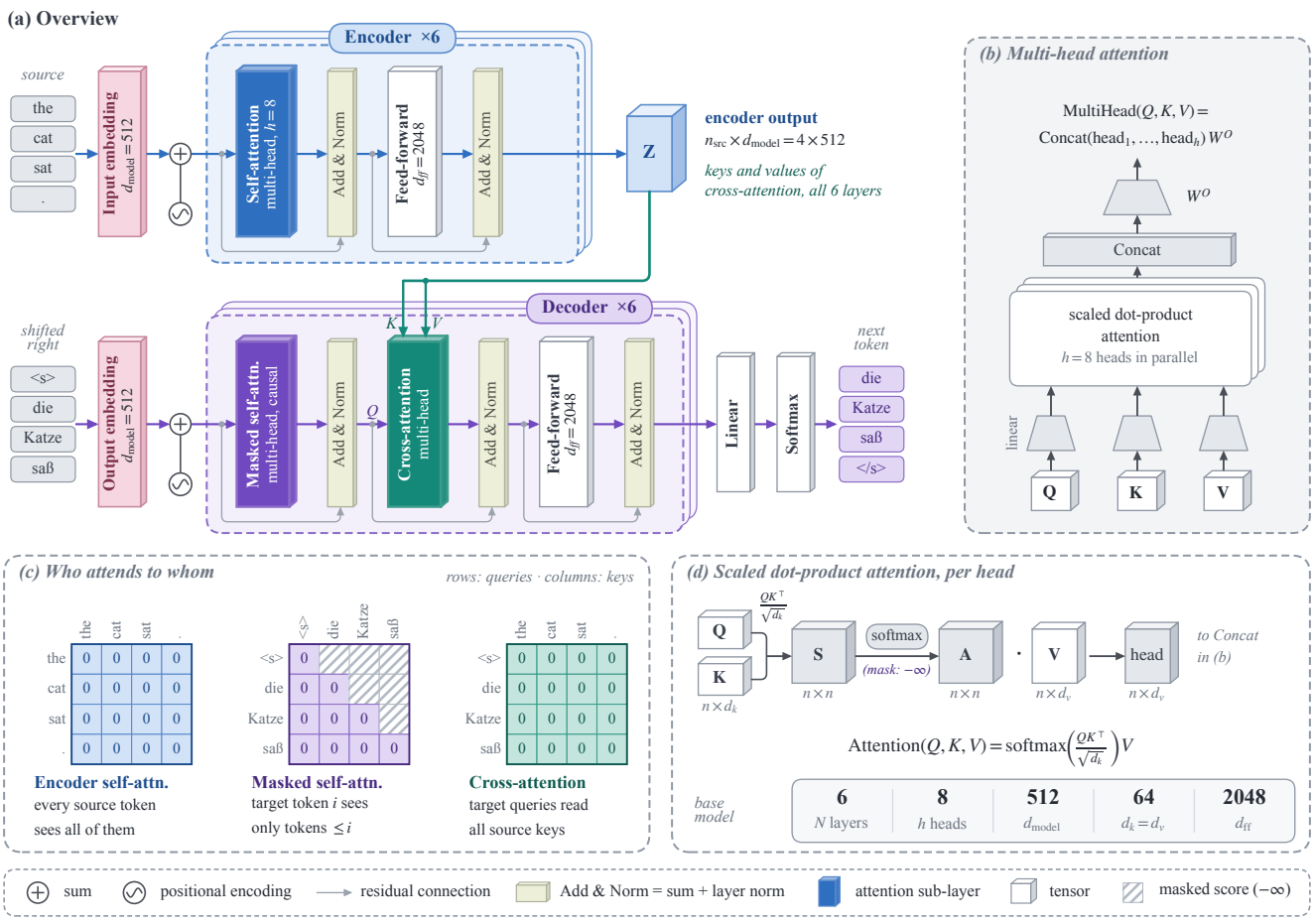

(a) Overview
source
the
cat
sat
.
Input embedding
$d_{\text{model}} = 512$
Encoder ×6
Self-attention
multi-head, $h = 8$
Add & Norm
Feed-forward
$d_{ff} = 2048$
Add & Norm
Z
encoder output
$n_{\text{src}} \times d_{\text{model}} = 4 \times 512$
keys and values of
cross-attention, all 6 layers
shifted
right
<s>
die
Katze
saß
Output embedding
$d_{\text{model}} = 512$
Decoder ×6
Masked self-attn.
multi-head, causal
Add & Norm
K
V
Q
Cross-attention
multi-head
Add & Norm
Feed-forward
$d_{ff} = 2048$
Add & Norm
Linear
Softmax
next
token
die
Katze
saß
</s>
(b) Multi-head attention
MultiHead$(Q, K, V) =$
Concat$(\text{head}_1, \ldots, \text{head}_h)\, W^O$
$W^O$
Concat
scaled dot-product
attention
$h = 8$ heads in parallel
linear
Q
K
V
(c) Who attends to whom
rows: queries · columns: keys
the
cat
sat
.
Encoder self-attn.
every source token
sees all of them
<s>
die
Katze
saß
Masked self-attn.
target token $i$ sees
only tokens $\leq i$
Cross-attention
target queries read
all source keys
(d) Scaled dot-product attention, per head
Q
K
$\frac{QK^\top}{\sqrt{d_k}}$
S
softmax
(mask: $-\infty$)
A
V
head
to Concat
in (b)
$n \times d_k$
$n \times n$
$n \times n$
$n \times d_v$
$n \times d_v$
Attention$(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$
base
model
6
N layers
8
h heads
512
$d_{\text{model}}$
64
$d_k = d_v$
2048
$d_{\text{ff}}$
sum
positional encoding
residual connection
Add & Norm = sum + layer norm
attention sub-layer
tensor
masked score $(-\infty)$